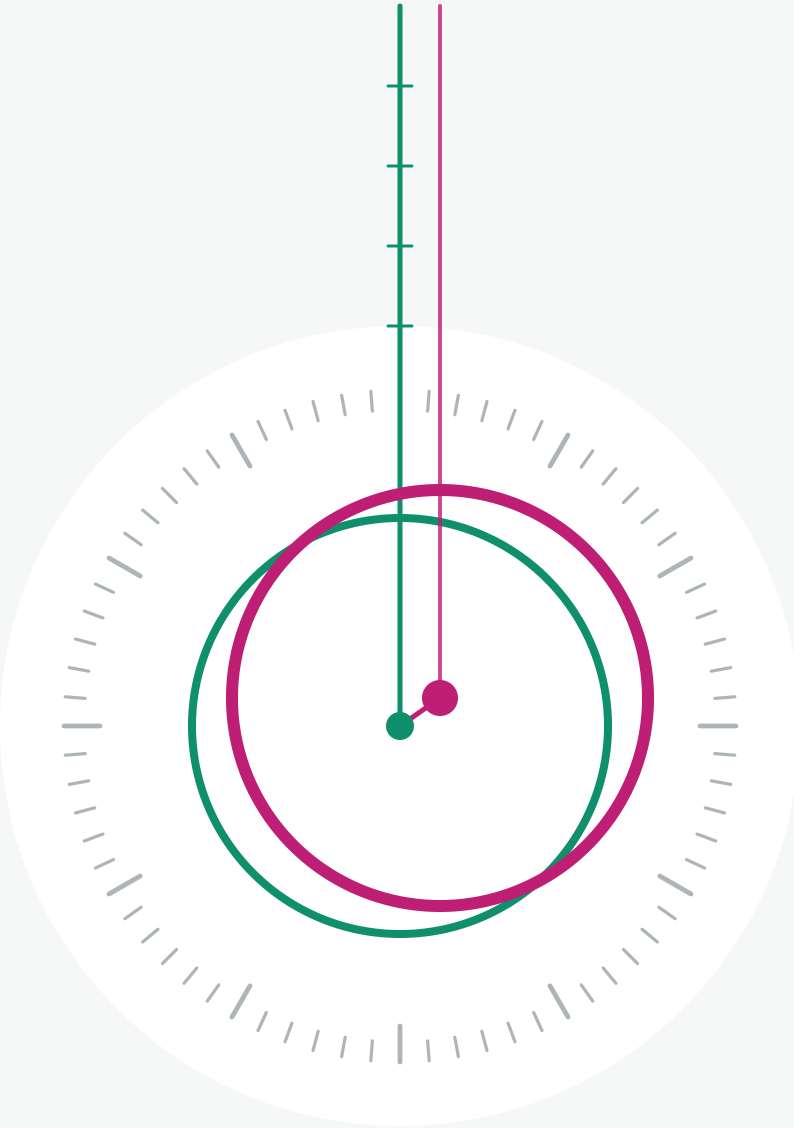
DATUM LINE

# Confidence with controls.

Every lab’s cells, on one map.



Kat Titterton, San Francisco



macrofluidic.me/resume

# Katherine Titterton

2023–2026 **Axiom**
founding scientist

2022–2023 **insitro**
neuroscience assays

2021–2022 **MIT**
expansion microscopy

2018–2021 **abbvie**
tau, neurons

2017–2018 MGH HARVARD MEDICAL SCHOOL
frog eggs

2017 **COLGATE** BA
neuroscience + math biology

## MY WORK

1. Cell Painting in primary human hepatocytes
   *Cell Systems* 2026, Ewald, Broad Institute
2. Counting cells predicts bioactivity benchmarks
   *Nat Commun* 2026, Seal, Univ. of Cambridge
3. Cell Painting of hepatotoxic botanicals
   *bioRxiv* 2026, Ollitrault, EMBL-EBI
4. An antibody to pathological tau
   *Mol Neurodegen Adv* 2026, Yanamandra, AbbVie
5. Capturing synaptic proteopathic seeds
   *iScience* 2026, Sebastian, AbbVie
6. A neuronal tau aggregation assay
   *J Biol Chem* 2025, Wu, AbbVie
7. Multiplexed imaging of brain nanostructures
   *Nat Commun* 2024, Kang, MIT
8. Tau aggregation in human iPSC neurons
   *iScience* 2022, Manos, AbbVie
9. Neurosteroid photolabeling of $GABA_A$ receptors
   *Eur J Med Chem* 2019, Wu, U Illinois Chicago

# Do my cells look “normal”?

asked @ five benches
… answered @ none

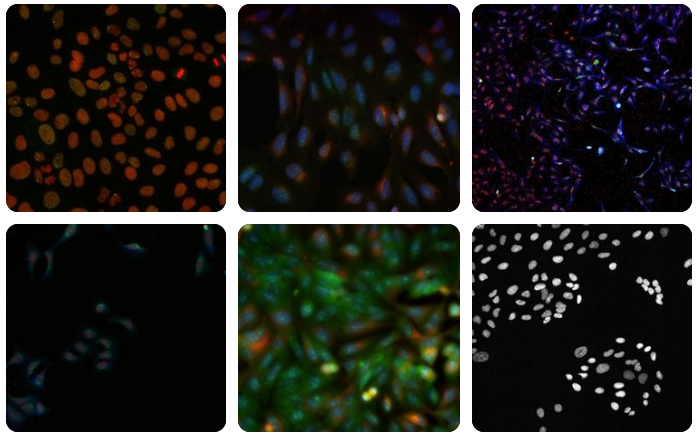
# Today each lab judges its own “normal”, alone.

$28B a year of US lab research others cannot reproduce [6]

36% ≈ $10B, cells and reagents | 64%, other causes

1
**search**

do my U2OS cells look “normal”

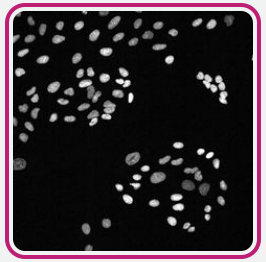


one cell line, six papers, all look different

2
**ask**

To: whoever ran this line last

Subject: does this look right to you?



screenshot.png

one screenshot, one opinion

3
**guess**

$ RUN IT ANYWAY

**hundreds of thousands**

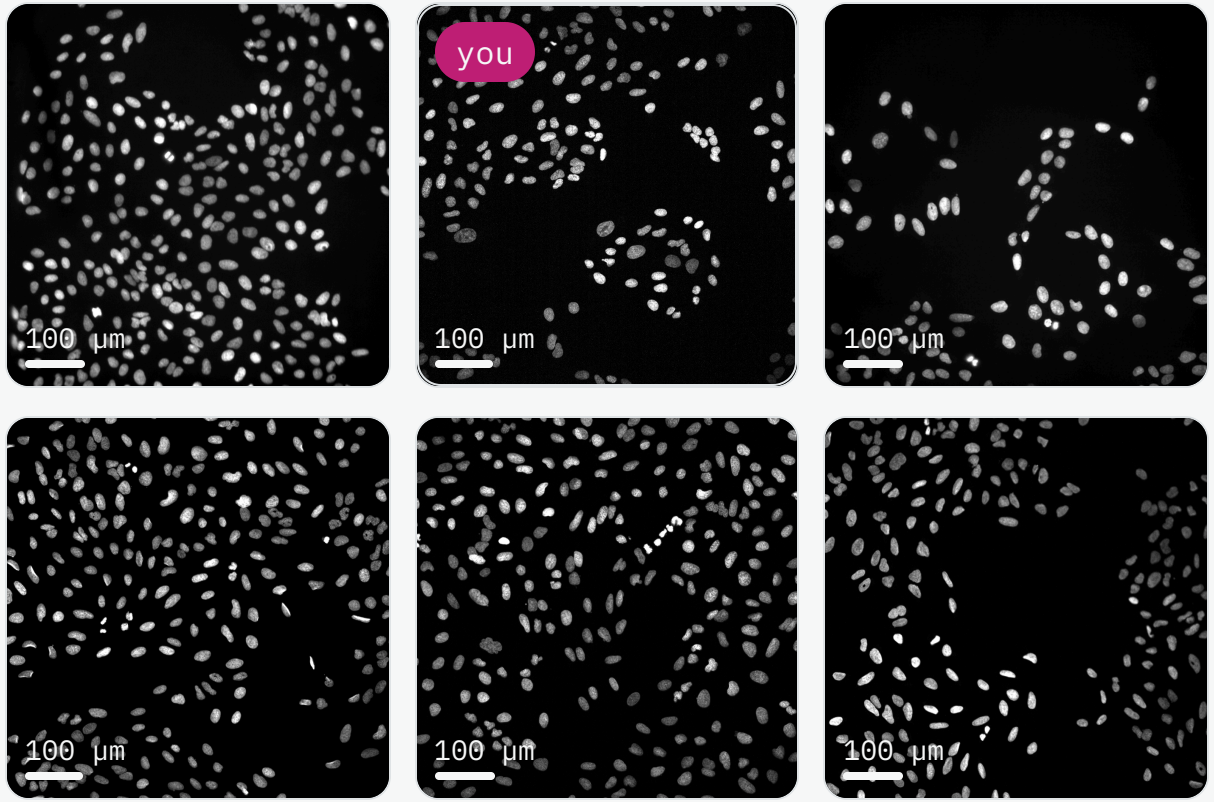
of R&D dollars at risk

WAIT TO BE SURE

**months**

of development lost

# One lab, or six?



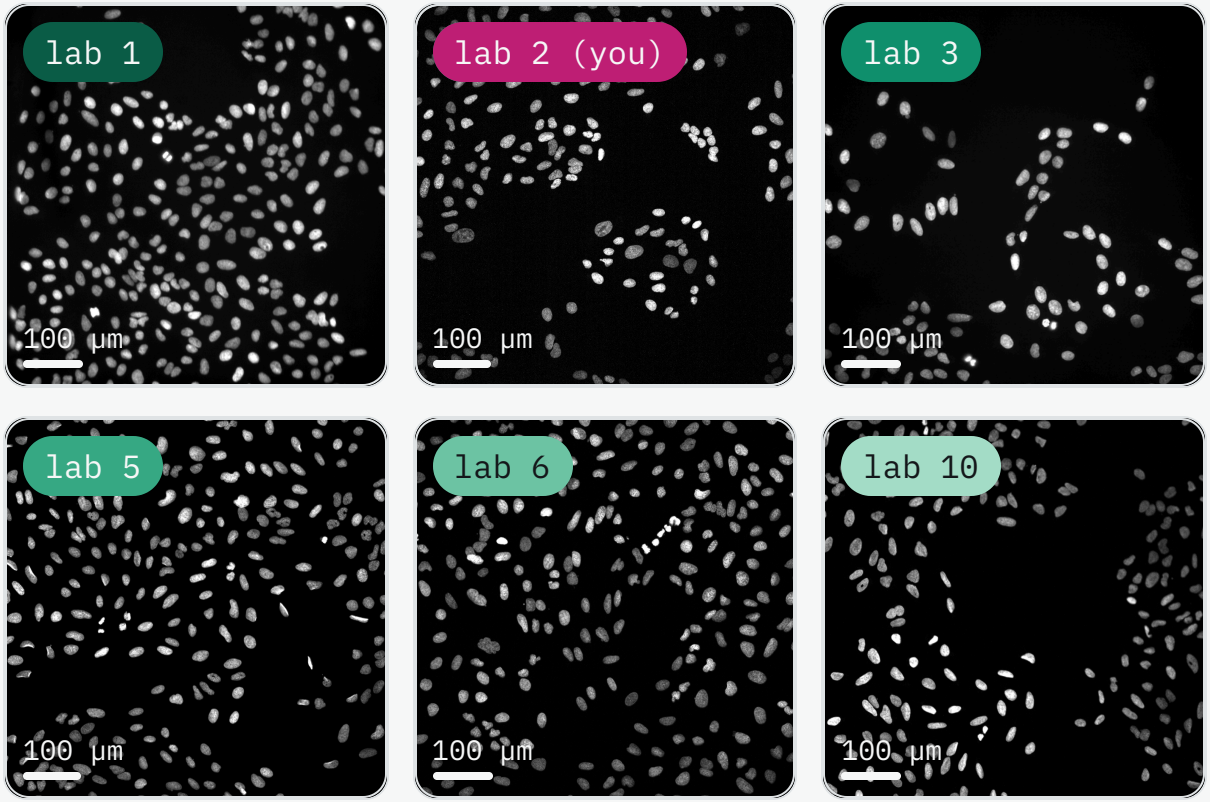

same cells ✓

same protocol ✓

no drug ✓

**fig 1** Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

# Could you tell? A classifier can.




**99.9%** names the right lab, of 11

**9%** a random guess

**fig 1** Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

# One well, onto the map.

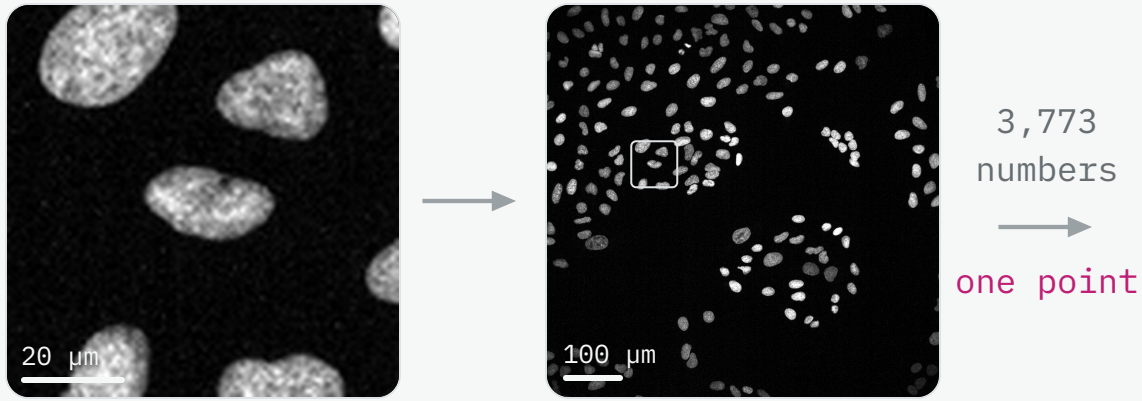






**fig 1** Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

# Eleven labs.
# Eleven “normals”.




**6×**

Lab to lab is 6× the spread of wells inside one plate.

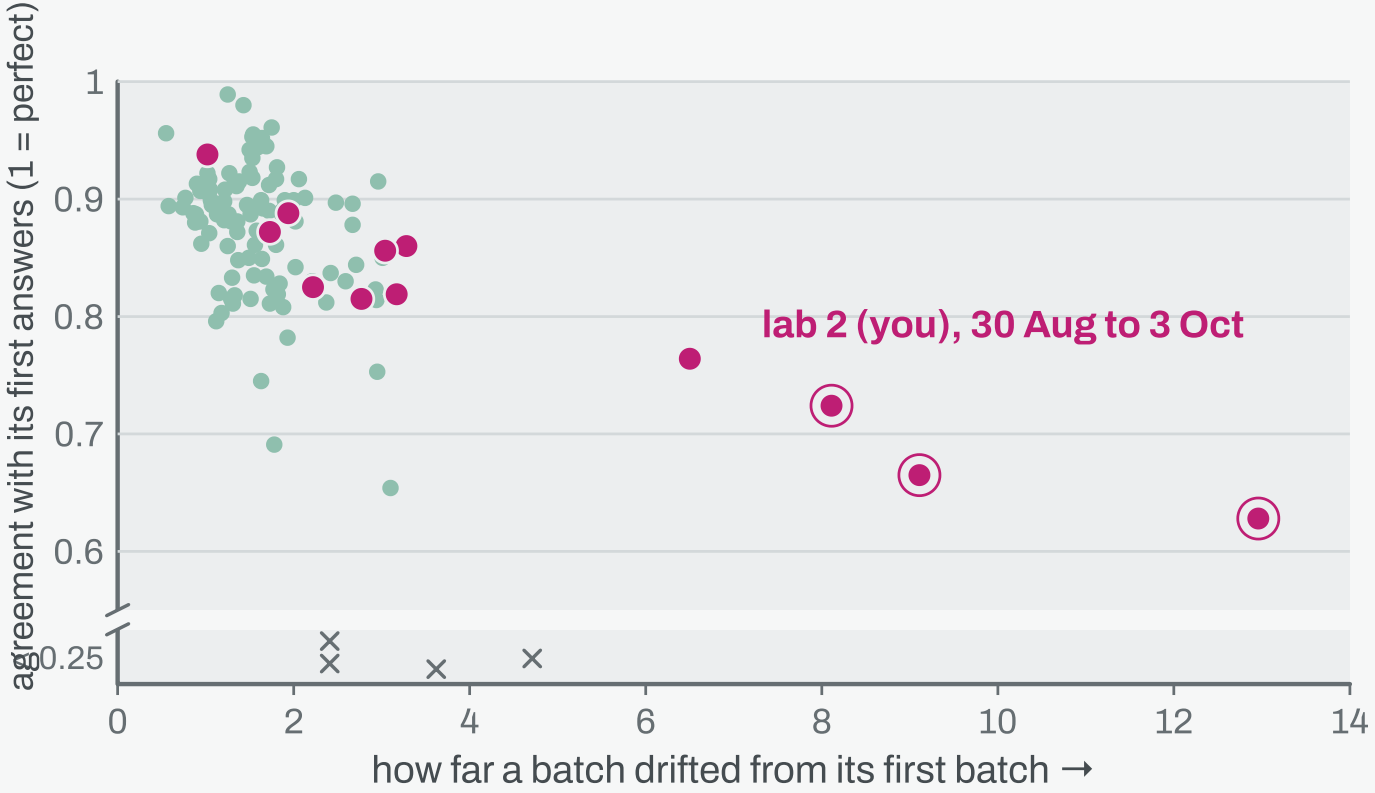
lab to lab

spread inside one plate

6.2 within-plate spreads, lab to lab (results/e13)

**fig 2** 15,352 DMSO wells, PCA of 3,773 numbers, JUMP [7], A2, A3, A13

# Where you sit predicts what you measure.




In every lab: more drift, less agreement. ρ = −0.71 within labs, each lab centred, 118 batches (results/e10)

## Lab 2 (you)

From 30 Aug to 3 Oct it drifted 3–5× its summer level.

**0.85 → 0.67**

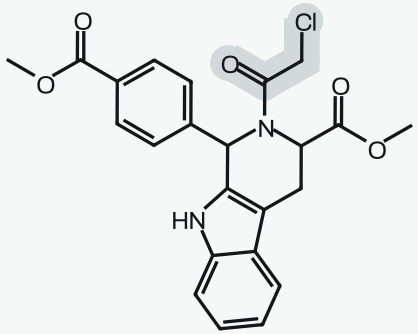
Same drugs, checked against its own June answers.

**fig 3** 118 batches, 11 labs, JUMP [7], A6, A8, A12, A13

# Does my control look like yours?

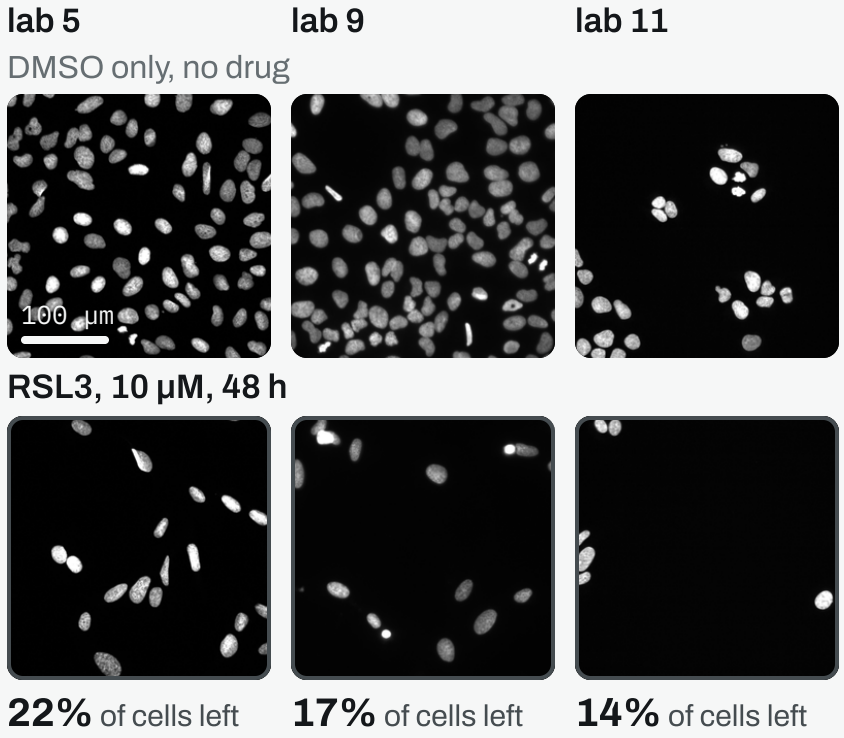
## RSL3

a common ferroptosis control,
in about 1,200 papers [16]




$C_{23}H_{21}ClN_2O_5$, as JUMP records it,
no stereo; active form (1S,3R) [19]

shaded group binds GPX4 [19]
↓ lipid damage builds up
↓ cells die: ferroptosis [20]




# 14–22%

of cells left in all three labs.
Same dose, same kill.

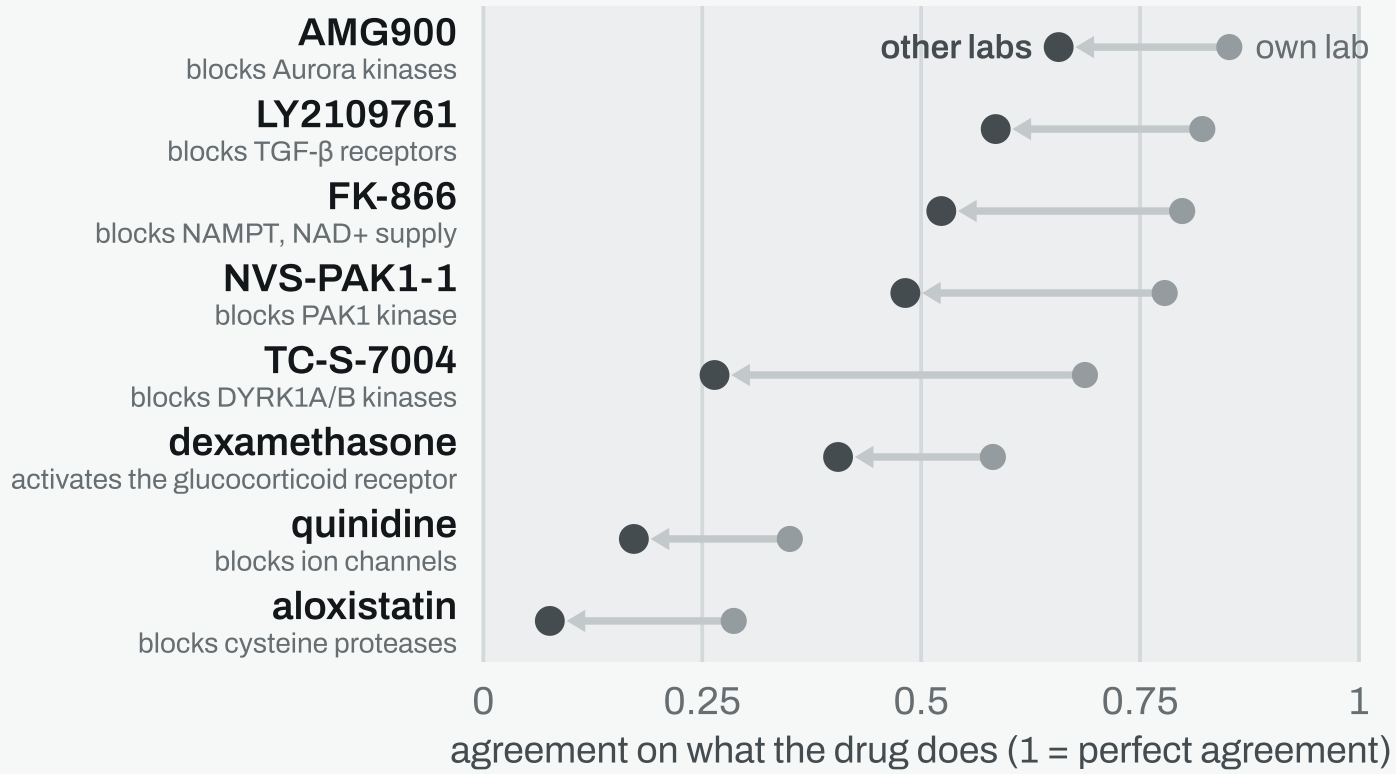
Yet the wells look far less alike across labs.

Well to well at 10 µM: 0.90 inside a lab → 0.49 across labs; known drugs on the same plates 0.74 → 0.26 (1 = perfect agreement, A9).

Seven wells in three labs: an example, not proof.

**fig 4a** RSL3 at JUMP's stated dose [7], one well per plate, % of cells left = nuclei per field against DMSO wells of the same plate, lab mean, lab 7 (0.625 µM, 80% left) left out, DNA stain, 300 µm, PubMed [16], A9

# Does my control look like yours?




## 0.64 → 0.40

All eight known drugs, 5 µM in every lab.
Plates agree less across labs than in one.

---

On the map, labs agree a little more:

## 107 of 125

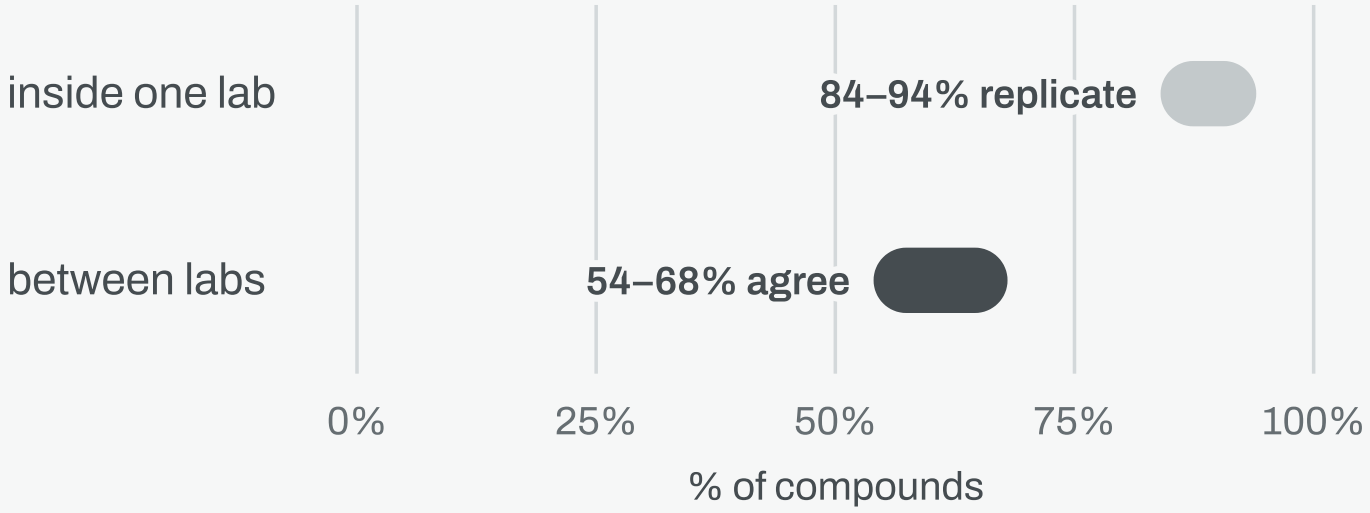
batches where labs agree more on the map than after each lab's own correction.

plate pairs, eight drugs (results/e8); per batch, 7 drugs: 0.75 raw → 0.76 usual fix → 0.78 on the map (results/e12)

**fig 4b** agreement between pairs of plates, same lab and across labs, 5 µM in every lab [7], 1,829 plates, 11 labs, targets [24], A9, A14

# Lab A + lab B ≠ one dataset.

Four labs, one protocol, the same cells, serum and dyes.




**“not directly comparable”, the authors wrote.** [13]

**$30M**

Roche paid it for one cell dataset. [11]

**Why now**

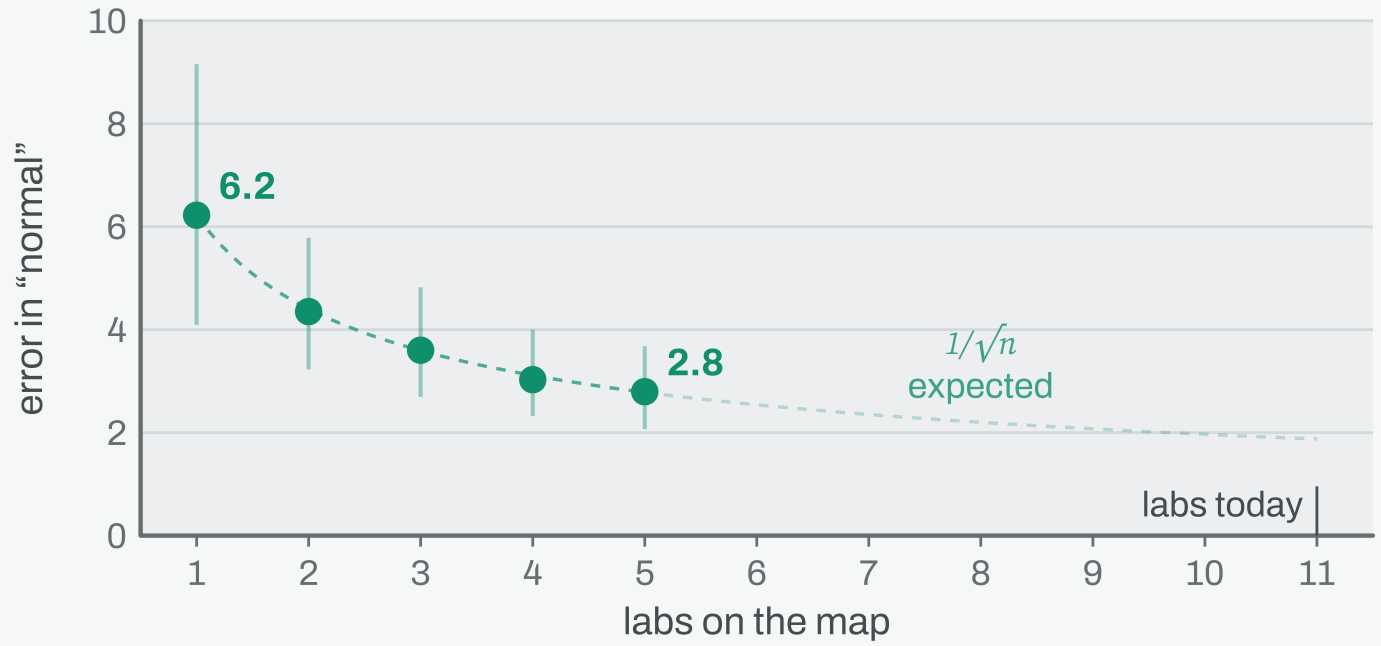
2023 the public reference: JUMP [7]

2025 the FDA moves away from animal tests [9]

now off-the-shelf models read cell images A10

**fig 5** EU-OPENSCREEN, one protocol, Hep G2, as reported [13], A23

# Every lab that joins sharpens “normal” for everyone.




**6.2 → 2.8**

The error in “normal”, with one lab and with five.

Map, today

↓

Certificate, next

↓

Subscriptions, then

↓

Journals, raw images, every paper

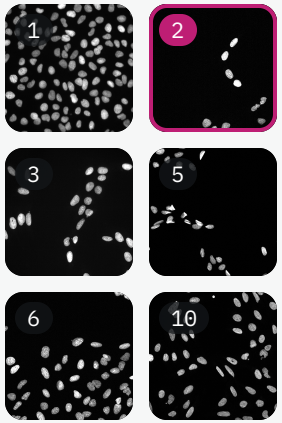
↓

…

**fig 6** error: distance between two maps built from different labs, in within-plate spreads, measured to five labs, dashed beyond, 1,871 plates, JUMP [7], results/e13

# One shared repository of raw cell images.

IN, FREE




control images from eleven labs (A18)

DATUM LINE

**one map**
every lab placed on it

**raw images**
kept, with their provenance

**certificate**
every batch read against the field

**benchmark**
every method graded on one test

93,228 control wells today

OUT, BY SUBSCRIPTION

**scientists**
is my batch “normal”? whose cells look like mine?

**AI agents**
comparable images with provenance, by API

**companies**
a certificate on cell data they buy

Labs keep their data and a share of every subscription. Proposed terms, A15, A24.

# 93,228

control wells from eleven labs, on the map today. results/e1

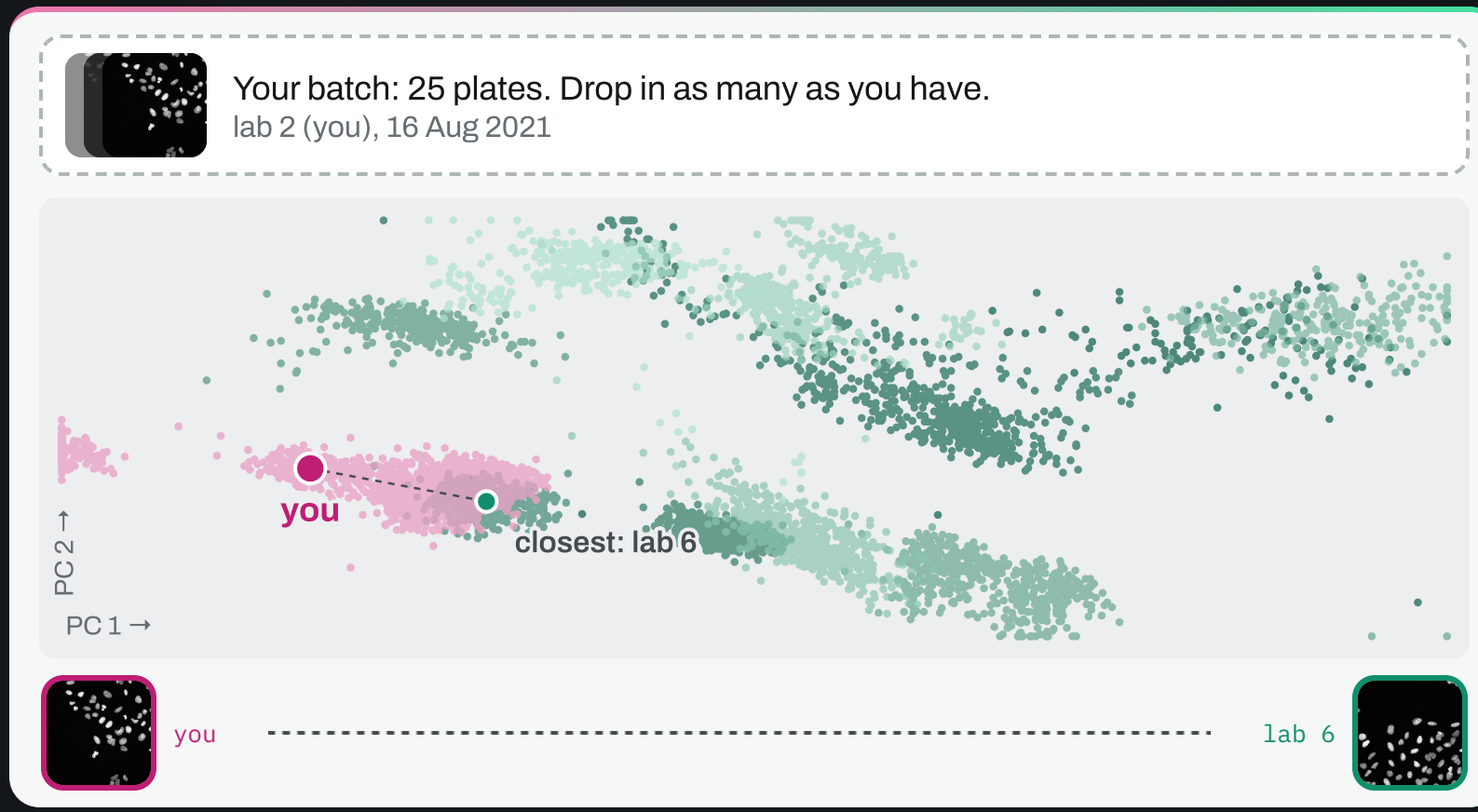
One subscription: the repository, the map, the benchmark.

Public archives keep finished datasets (656 TB, free [17]). This keeps every lab’s live controls, aligned and checked.

Every cell result becomes checkable against the field.

the product as proposed, labs join free, subscribers pay for access, counts from results/e1, JUMP [7], A15, A18, A24

# Is my batch “normal”, next to every other lab?




## Your batch, answered

**Do my controls look “normal”?**

yes, within the usual range

**Which lab is closest to mine?**

lab 6 4.4 away

**Do my drugs match other labs?**

closer on the map: 0.70 → **0.77**

**Did my cells grow as usual?**

yes: 188 per well, in range

The four questions it answers first (proposed).

**fig 7** replayed on public data, JUMP [7], A11, A18

# Which labs make the most reliable data?

LEADERBOARD, U2OS

controls only

| rank | lab | batch | plates | field score |
|---|---|---|---|---|
| #1 | lab 5 | 28 Jun | 7 | 82 |
| #2 | lab 5 | 2 Jul | 6 | 79 |
| #3 | lab 5 | 6 Jul | 5 | 79 |
| #4 | lab 5 | 16 Jul | 8 | 78 |
| #5 | lab 5 | 1 Oct | 6 | 77 |
| … | | | | |
| #48 | **lab 2** | 2 Aug | 18 | your best 58 |
| … | | | | |
| #95 | **lab 2** | 16 Aug | 25 | this batch 36 |

## Your standing

**#95** of 129 batches

### Batches in a row on the map

best in the field: 33

### To climb: steadier batches

you 2.9

field 1.2

drift, lower is better

Only you see your rank, unless you share it.

**fig 8** score = 100 − mean percentile, DMSO wells, JUMP [7], A17

# Your certified data, shared by subscription.

**U2OS controls, shared by 11 labs**

| lab | plates | batches certified |
|---|---|---|
| lab 3 | 237 | ■■■■■■□□□□□■■ |
| lab 6 | 225 | ■■■■■■■■■■■■ |
| lab 2 (you) | 219 | ■■■■■■■■■■□□□ |
| lab 10 | 205 | ■■■■■■■■■■■■■■■■■■■■ |
| lab 8 | 199 | ■■■■ |
| lab 5 | 197 | ■■■■■■■■■■■■■■■■■■■

# Six questions, six tools. Use all of them.

Each answers one question well.
Each misses something, Datum Line too.

| tool | answers | does not see |
|---|---|---|
| **DNA fingerprint (STR)** | Is it the line the label says? | how the cells behave A7 |
| **vendor certificate** | Was it fine when it shipped? | anything after shipping |
| **plate controls** | Did this plate work? | a whole lab drifting A1 |
| **in-house platforms** | Is my own data consistent? | lab differences, removed on purpose A3 |
| **data factories** | Where do I get cell data at scale? | other labs' "normal" A23 |
| **Datum Line** | **Do my cells look "normal", next to other labs?** | which cell line it is; unstained images, not yet A7, A10 |

# Eighteen months, four milestones.

## $1.8M

$0.9M line by line, ×2 for the usual overrun A21

→ seed round

### 1

| | widefield | confocal |
|---|---|---|
| U2OS | ■ | ■ |
| A549 | ☐ | ☐ |
| HepG2 | ☐ | ☐ |
| iPSC | ☐ | ☐ |

**benchmark playground**

+ known drugs + planted faults A4, A16

### 2

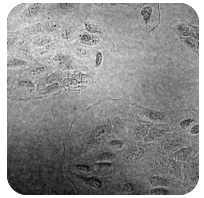
**11 + 3**

labs today + 3 partners

**three partner labs on the map**

U2OS first, then any line

### 3



**99.5%**

right lab today, 8 labs

**model for any image**

from one unstained image A10, A19

### 4

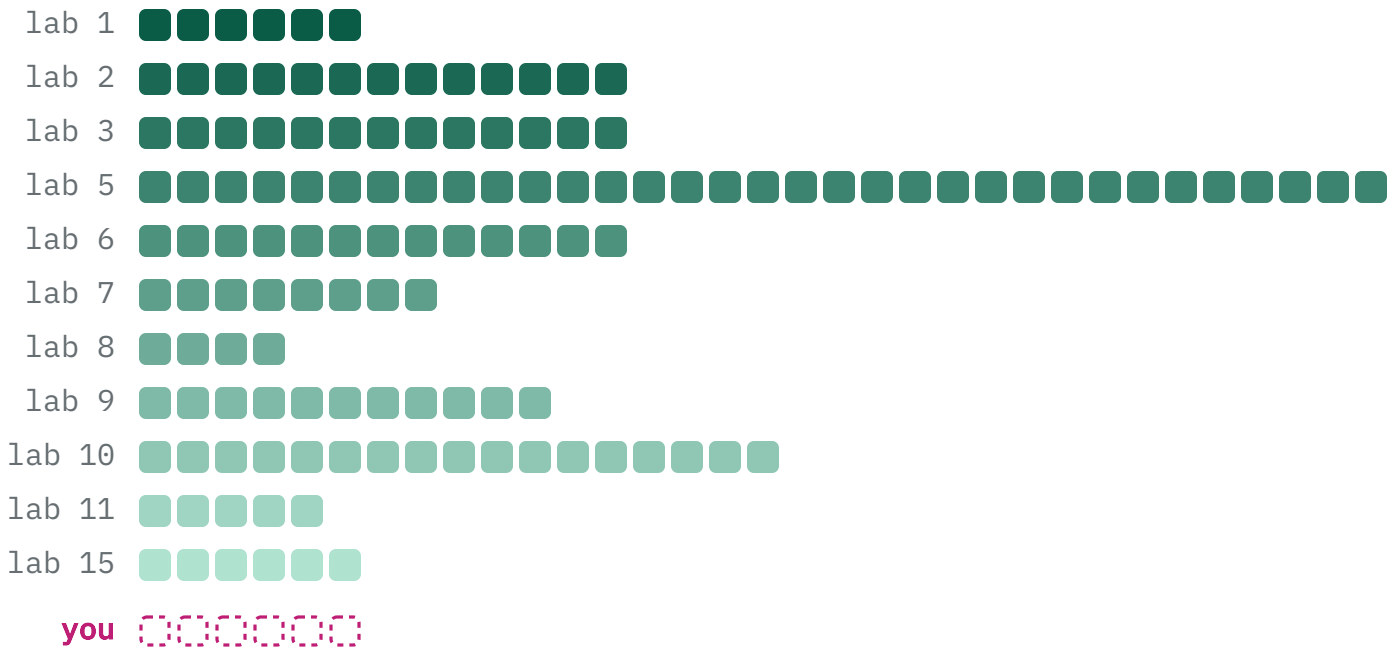
**first paid certificate**

a company pays ~$10–50k to check cell data it bought A15

proposed, milestones in order, not dated

# Every plate you image, placed on the map.

Labs share control images, free.
Each sees its “normal” next to everyone else’s.




## $1.8M

raising, pre-seed, eighteen months A21

**leader**
to run it

**bench**
3 partner labs

**partners**
2 for the benchmark

**Have control images?**
**Talk to me.**

**fig 10** 1 square = 1 batch, 1,871 plates, JUMP [7], A15, A20, A22, A24

# The rest, if you ask.

Every number comes from the code here or a cited source.
Unfinished work is marked pending.

A1 **Why not trust the plate's controls?**
sick plate, sick control

A2 **What is the math?**
six calculations on 3,773 numbers

A3 **Why not batch-correct it away?**
erases most of the lab fingerprint

A4 **Is the model the moat?**
counting cells often wins

A5 **Why not just count cells?**
same count, different texture

A6 **How often do I re-certify?**
your "normal" expires

A7 **Isn't this cell-line DNA testing (STR)?**
which cells, how they behave

A8 **Is the assay itself working?**
known answer

A9 **Does it change the result?**
same drug, different answer

A10 **Does it work without stains?**
bet on brightfield

A11 **Does it flag too often?**
rarely, six of 129 batches

A12 **What moved?**
stain brightness, not cell count

A13 **How does one lab move over time?**
own "normal" fine, other labs: moved

A14 **What does the map buy?**
same drug, closer between labs

A15 **Who pays?**
labs join free, companies pay

A16 **What would the money build?**
ugliest dataset, on purpose

A17 **Is the leaderboard fair?**
ranked on controls, never on results

A18 **What is private?**
control wells leave, nothing else does

A19 **What is the machine learning?**
image in, position and correction out

A20 **What about biosecurity?**
cells show their lab, labels can lie

A21 **How much, and what for?**
planned $0.9M × 2 = $1.8M asked

A22 **How big?**
count × price, three streams

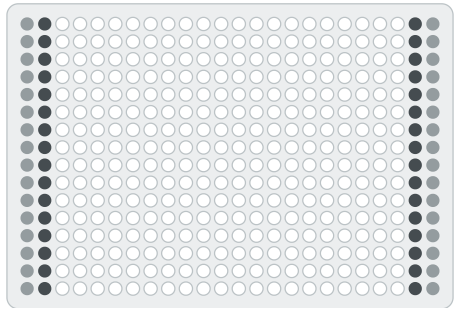
A23 **What is cell data worth?**
priced in deals, labs disagree

A24 **Why would a lab share?**
owned by the lab, verified by anyone

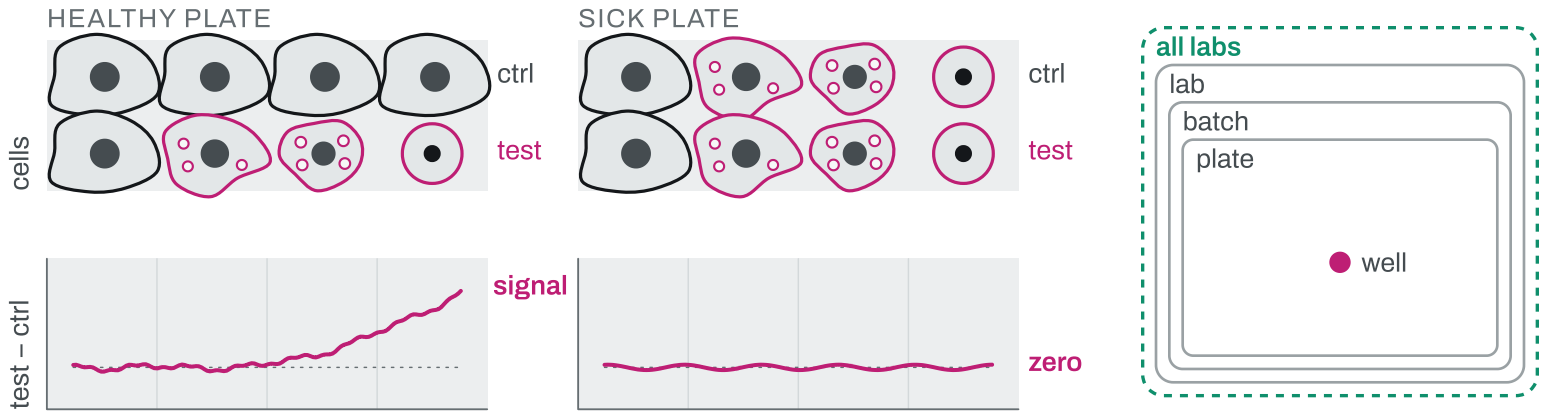
# If the plate is sick, its controls are sick too.

Controls sit on the same plate as the tests.

A REAL JUMP PLATE, 384 WELLS



DMSO 32 known effect 32
test 320




*test* = *"normal"* + *plate* + *effect*

*control* = *"normal"* + *plate*

*difference* = *effect on this plate*

plate cancels from the number, never from the cells

WHAT CAN SEE A SICK PLATE?

A view from outside the plate.
Same cells in other labs: the map.

**Fig. A1** left: JUMP plate 1053601756, lab 2, right: schematic, no measurements

# Every result is one of six plain calculations.

Results compare averages, or test a model on unseen plates.
Drug discovery removes the lab; a quality check reads it.

### 1, IMAGE TO NUMBERS [1]

$$e = \varphi(image)$$

Three ways to turn images into numbers gave similar results on liver cells.
Tool is not the asset.

### 2, WHICH LAB? 99.9%

$$f : e \to lab$$

Names the lab from one well, on plates it never saw.
Far above chance (9%): each lab leaves a fingerprint.

### 3, HOW FAR FROM "NORMAL"? [e6]

$$d = \| E[e \mid plate] - E[e \mid all\ labs,\ c] \|_{\sigma}$$

How far this plate's average sits from all labs' average, in units of usual spread.
Direction says what moved.

### 4, SOURCES OF VARIATION pending

$$Var(e) = V_{model} + V_{lab} + V_{batch} + V_{plate} + V_{well}$$

Splits variation into microscope type, lab, batch, plate, well.
Shares add up to 100%.

### 5, REMOVE KNOWN CAUSES pending

$$e_{\perp} = e - \hat{E}[e \mid c]$$

Remove only causes known to be the instrument.
Subtracting each plate's controls goes further: right lab only 24% of the time.

### 6, DID THE DRUG CHANGE THE CELLS?

DISTANCE $\delta = b(\|e - E[e \mid DMSO]\|)$

CLASSIFIER $g : e' \to label$

Distance from the DMSO average, capped so one odd well cannot dominate.
Or a model, after aligning to the reference.

# The usual correction erases most of the lab's fingerprint.

What counts as "normal" depends on every variable below.
JUMP records the left column; nobody records the middle.

## RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

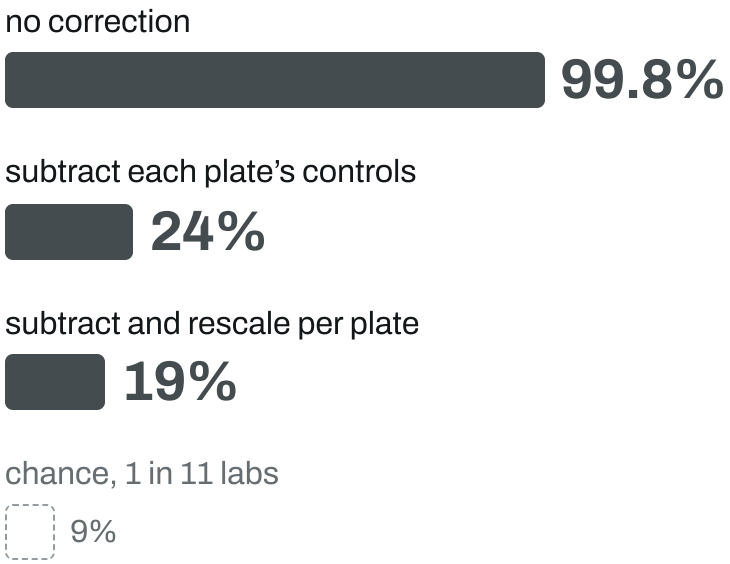
in every image file too:
image size, stored bit depth

## NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity
- compound dose

not required to publish today
in our own data: tracked, every one

## CAN A MODEL TELL WHICH LAB?
## HIGHER = CLEARER LAB FINGERPRINT




29,943 DMSO wells, 16 per plate
tested on plates it never saw

Batch correction is built to remove this signal [18], variables: JUMP metadata [7]

nature communications, 2026, [2]

# On weak benchmarks, counting cells often wins.

I coauthored this paper.
Benchmarks decide which model looks good.

… counting cells achieves unexpectedly high performance in these benchmarks, making them less useful for discerning whether additional properties, such as phenotypic profiles (mRNA or Cell Painting), provide additional useful information on bioactivity. … we recommend filtering benchmarks to exclude such assays and including a cell-count baseline. … we confirm that models leveraging Cell Painting image-based profiles outperformed the baseline cell count model. …

ABSTRACT EXCERPT, VERBATIM finding recommendation, what held up

## 209 and 270

tests in two popular benchmarks
many really measure cell health

## Images win

on 24 tests aimed at one protein each
Cell Painting beat the cell count

WHY IT MATTERS HERE

Whoever sets the baseline sets the benchmark.

benchmark playground, a cell-count baseline on every task, planted faults, every model scored, A16

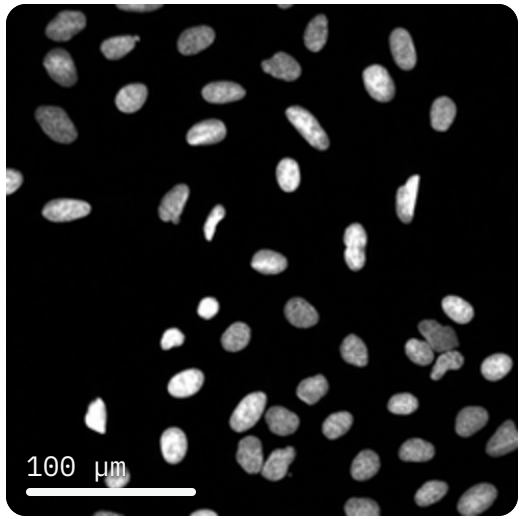
Seal S, Dee W, … **Titterton K**, … Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*, 2026.

# Same cell count, different texture: only images see it.

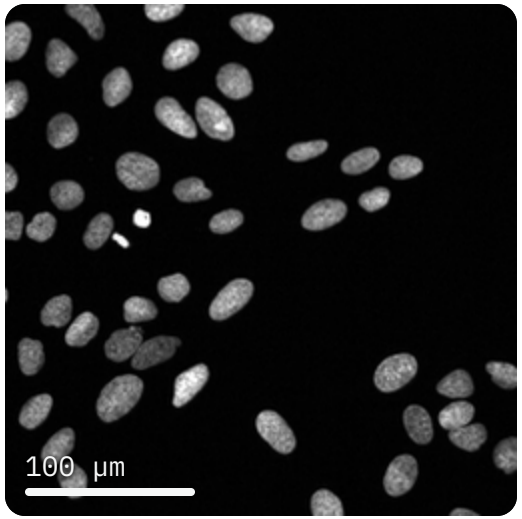
Count moves only when cells die or divide.
Image moves when anything changes.

SAME CELL LINE, SAME PROTOCOL, SAME MICROSCOPE

each lab's typical plate




LAB 5 CV8000, confocal




LAB 6 CV8000, confocal

CELLS COUNTED

**196 ≈ 222**

gap 0.4× the usual spread → same

FINE DETAIL, % OF DNA SIGNAL

**0.94% ≠ 1.10%**

gap 2.4× the usual spread → different

fine detail: finer than 5 µm, inside nuclei
usual spread: plate to plate, within one lab

NOT ONE LUCKY PAIR

All 44,325 plate pairs, lab 5 × lab 6:
texture differs in 91%, count in 33%.

TWO WAYS TO READ IT, A2

Classifier: does it belong with past runs?
Distance: how far out, which way back?

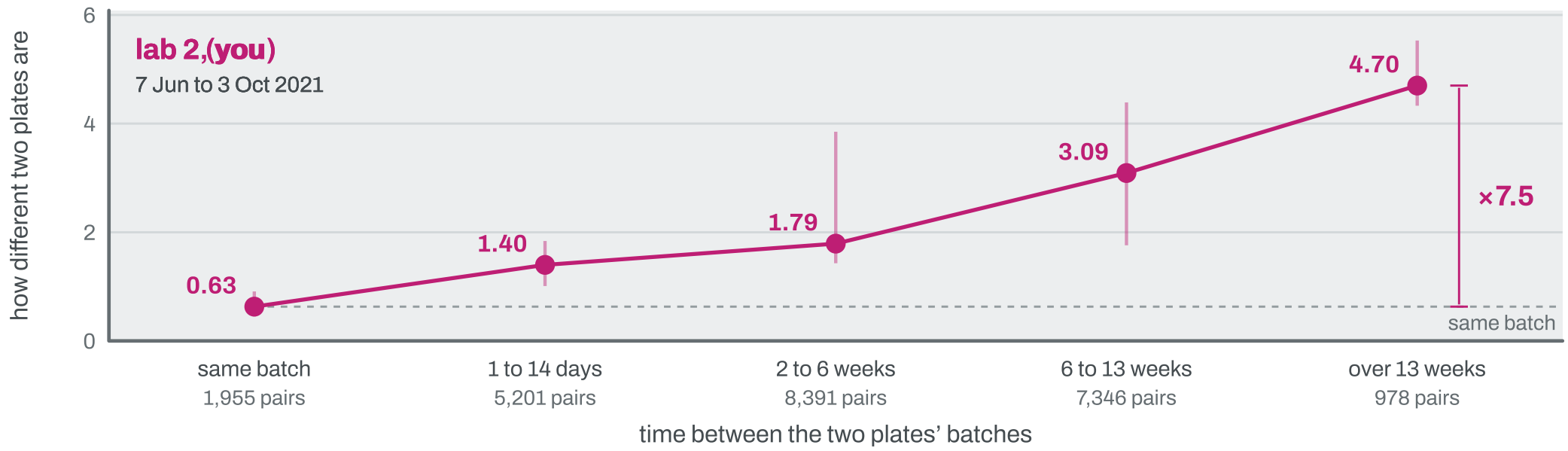
BOTH NEED A REFERENCE

Somebody else's images.
That is the map.

**Fig. A5** DNA channel, 300 µm squares, JUMP cpg0016 [7], CC0, nuclei by Otsu, both ways: published [1], our tuning unpublished

# Plates drift within two weeks: re-check every batch.

Two plates from one batch differ by 0.63; two weeks apart, 1.4. The jump past 13 weeks is lab 2's autumn move (A13).




IS THE CELL LINE RUINED?

This data shows when, not why.
What moved in lab 2: A12.

HOW FAST?

Within 2 weeks: ×2.2 vs same batch.
6 to 13 weeks: ×4.9.

HOW SOLID IS THE NUMBER?

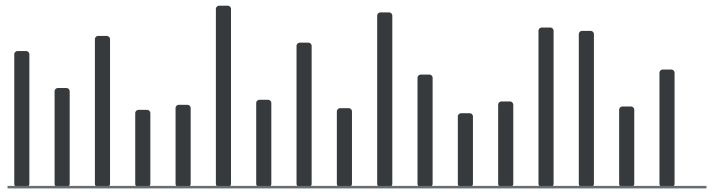
One lab, 219 plates, 13 batches.
Other labs not yet measured.

dot = median pair of plates, bar = middle half of pairs, 1 = how far a typical DMSO well sits from its plate's average, JUMP [7]

# STR says which line; images say where it was made.

One cell line, U2OS, in all 11 labs.
Images still name the lab: 99.9% right.

## STR, DNA FINGERPRINT

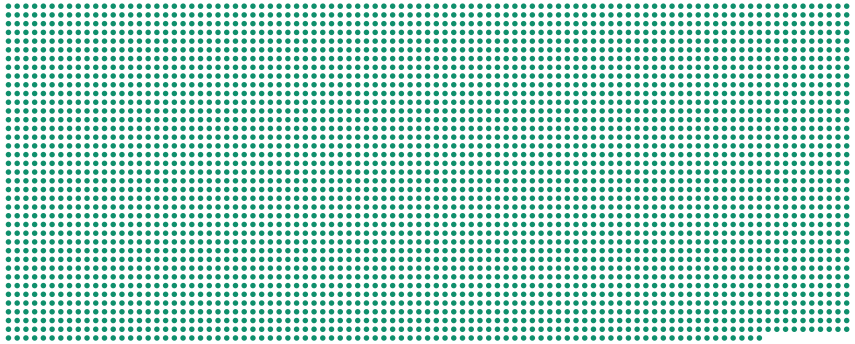


**17**
DNA markers

**periodic**
per cell line [14]

Standard test for identity.
Cannot say where or how the cells were grown.

## DATUM LINE



**3,773**
numbers per well

**every batch**
images already taken

Shows where cells were made, next to other labs.
Cannot say which cell line it is.

identity, STR + behaviour, Datum Line = **one certificate**

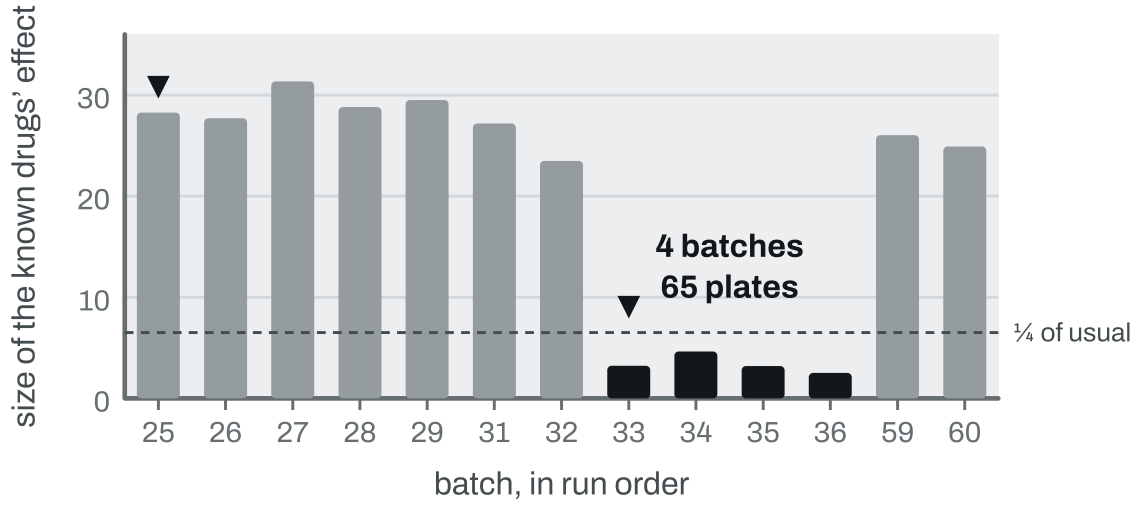
natural first partner: STR testing services

STR: 17 loci + amelogenin, ATCC [14], Datum Line: 93,228 U2OS DMSO wells, 1,872 plates, 11 labs, plates held out, JUMP [7]

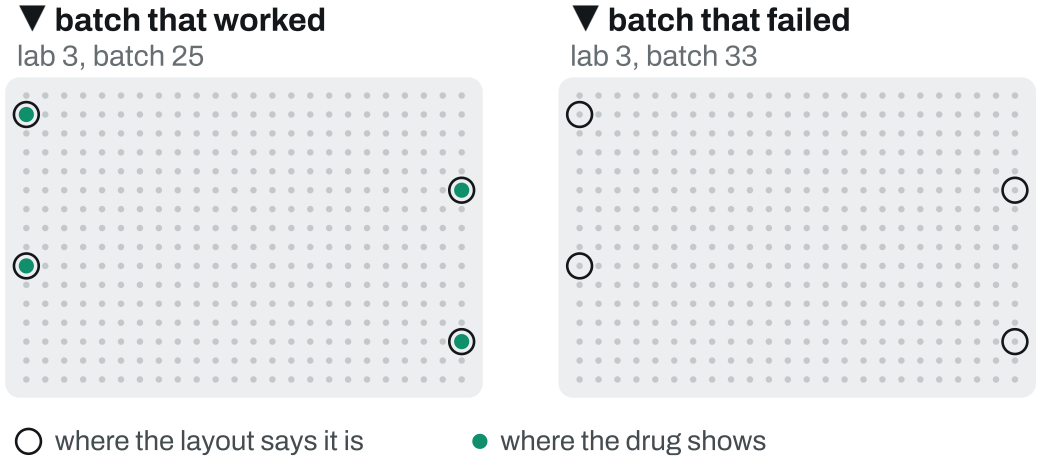
# Known drugs show whether the test itself worked.

DMSO wells show where a lab sits.
Lab 3: 4 batches where the known drugs did not work.

LAB 3, DID THE KNOWN DRUGS WORK?




WHERE DOES THE KNOWN DRUG SHOW?




**125 of 129**
PUBLIC BATCHES
KNOWN DRUGS WORKED

**1 of 4**
FAILED BATCHES CAUGHT BY
DMSO WELLS ALONE → USE BOTH

**–0.71 → –0.70**
FIG 3 LINK, DRIFT VS ANSWERS
WITHOUT THESE 4 BATCHES: HOLDS

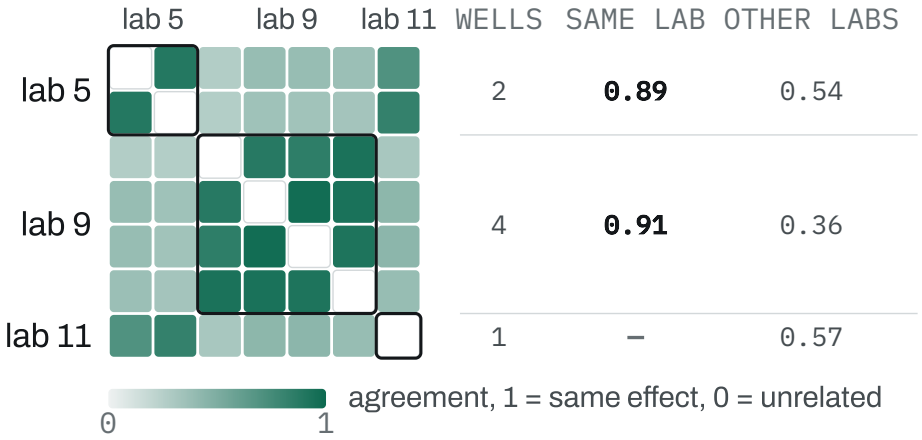
failed = under ¼ of the lab's usual effect + unlike other labs, 1,829 plates, maps: plates C13443aW, A12069aW, JUMP layout [7]

# The same drug looks less alike in another lab.

RSL3 at 10 µM from fig 4a, every well against every other.
Darker = more alike.

WHAT A DRUG SCREEN REPORTS — drug wells – DMSO wells, same plate

RSL3 AT 10 µM, EVERY WELL AGAINST EVERY OTHER




| | WELLS | SAME LAB | OTHER LABS |
|---|---|---|---|
| lab 5 | 2 | **0.89** | 0.54 |
| lab 9 | 4 | **0.91** | 0.36 |
| lab 11 | 1 | – | 0.57 |

SAME 7 PLATES, EACH WELL AGAINST ITS OWN PLATE'S DMSO

| | SAME LAB | OTHER LABS |
|---|---|---|
| RSL3, 10 µM | **0.90** | 0.49 |
| 8 known drugs, 5 µM, median | **0.74** | 0.26 |
| DMSO vs DMSO | **0.00** | 0.00 |

DOES IT CHANGE THE NUMBER?

Which lab ran it explains 20% to 71% of how strong a drug's effect looks.

HOW SOLID?

Seven wells: lab 5 two, lab 9 four, lab 11 one. Lab 11 has no same-lab pair.
Most of the gap sits beside lab 9, on 1536-well plates: 0.33–0.39 against 0.75.
Its known drugs split the same way.

LAB 7, LEFT OUT

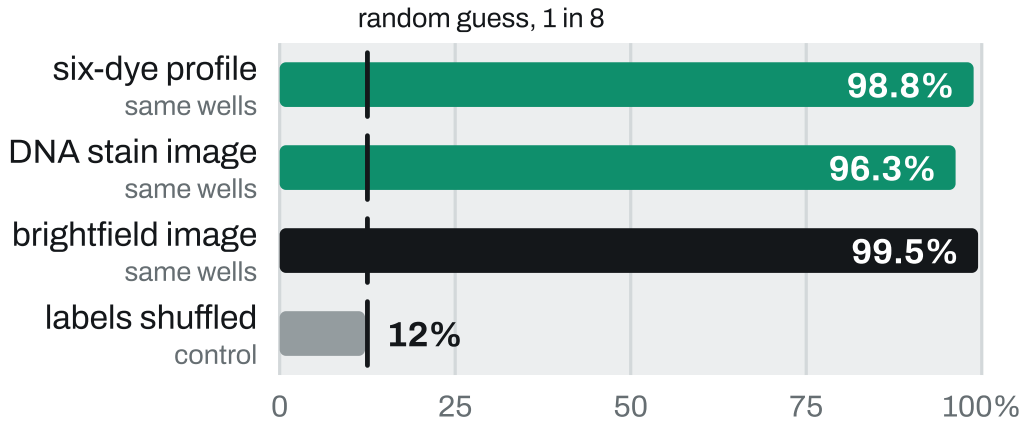
Dosed at 0.625 µM, a sixteenth [7]: RSL3 left 75–87% of its cells.
AMG900 at 5 µM left 36% there, 28–30% elsewhere: its cells respond.

agreement = cosine of two wells' effects, each well minus its own plate's DMSO wells, 8 drugs at full scale: 1,829 plates, 11 labs, JUMP [7]

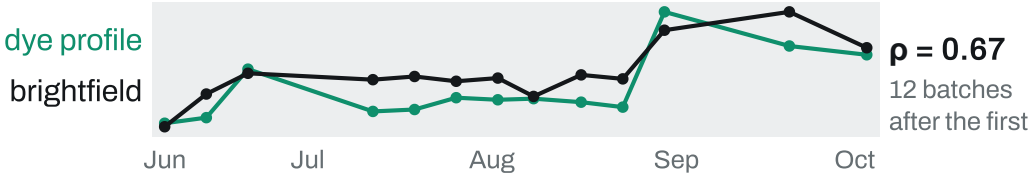
# One unstained image already names the lab.

Evidence: cells stained with six dyes.
Product: plain light, no dye.

CAN ONE WELL TELL WHICH OF 8 LABS IT CAME FROM?




DOES BRIGHTFIELD SEE LAB 2 DRIFT?, EACH LINE SCALED TO ITS PEAK

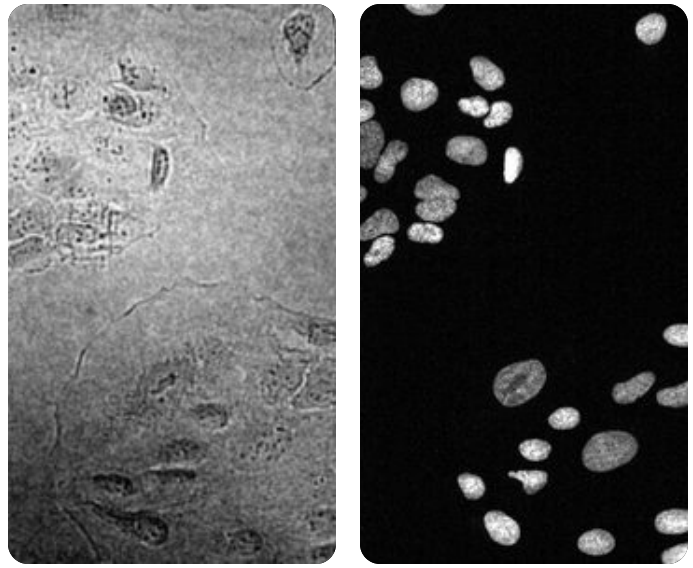



Not yet the whole map: beyond each lab's usual neighbour, brightfield and dye profile agree batch by batch no more than luck (45 of 102 against 41).

WHY THE BET IS FAIR

## No dye: fewer things can go wrong

~~stain lot~~ ~~dye lot~~ ~~staining step~~ A3



BRIGHTFIELD DNA STAIN

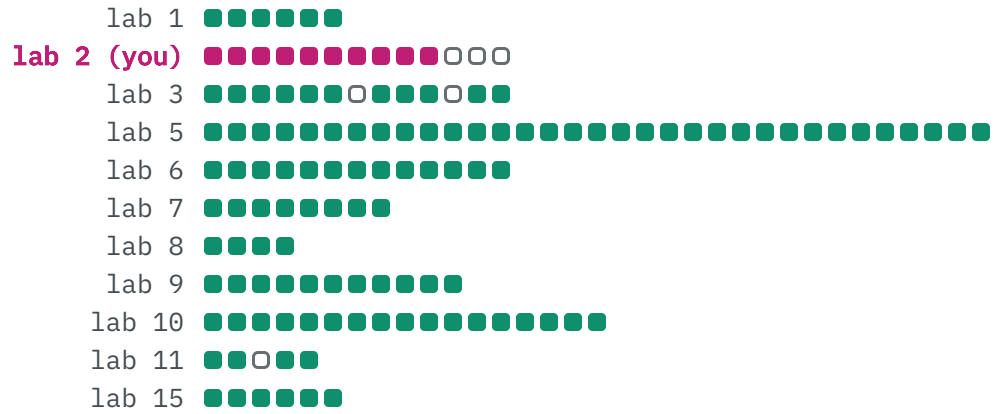
same spot in one well, lab 2

% named right, each lab counted equally, unseen batches, one DMSO well per plate, 1,338 plates, the same wells and model for every row, median of 20 splits of the batches, profile without its 8 frame-size columns, shuffled: 5 seeds per input, image model never trained on cells, JUMP [7]

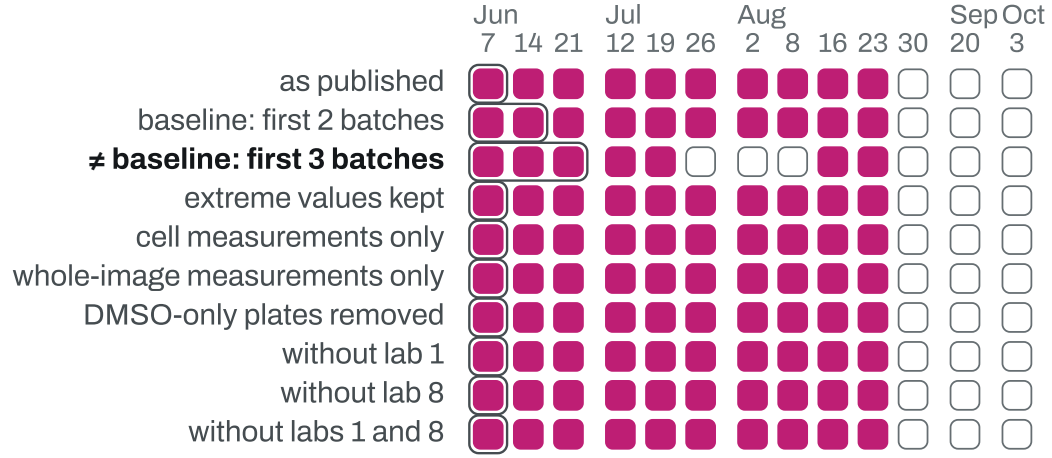
# It flags rarely: six of 129 batches.

8 of 11 labs never flagged.
The one way that differs flags more, not less.

ALL 11 LABS, 129 BATCHES




LAB 2 (YOU), 13 BATCHES × 10 WAYS OF CHECKING




lab 2 (you) · other labs · flagged · baseline: the batches it is compared with

WHAT A FLAG MEANS

Batch nearer another lab than its own “normal”, its first batch

LAB 2 (YOU), PLATES FLAGGED

**25 of 25** in flagged batches

**2 of 194** in the other batches

THE ONE WAY THAT DIFFERS

**+3 flags, 26 Jul → 8 Aug**

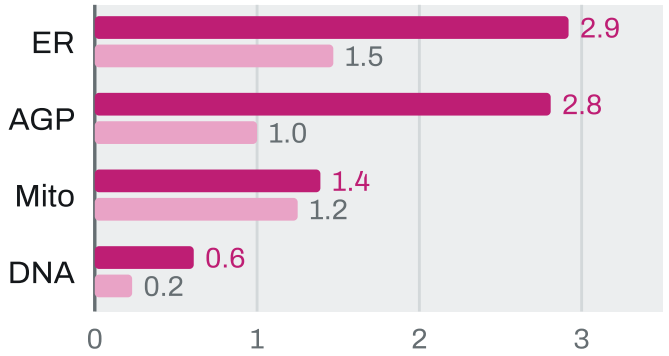
Its baseline takes in 21 Jun:
150 cells per well, lab 2’s lowest, A13

Each plate checked against its own lab’s first batch, 1,871 plates, 11 labs, JUMP [7], lab 2 over time: A13

# Stain brightness rose; the cell count did not.

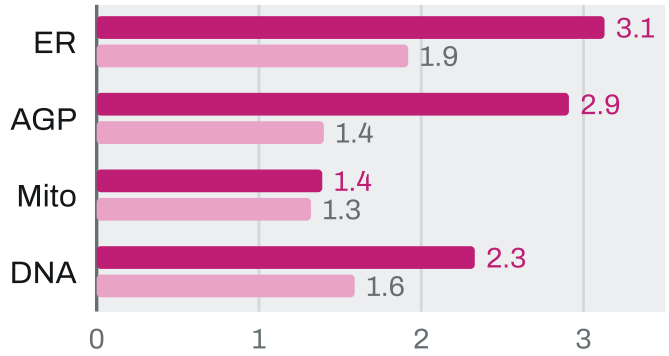
Lab 2 (you): flagged batches against its own “normal”, its first batch (7 Jun).

BRIGHTNESS




shift from 7 Jun, 0 = none, 1 = usual spread across labs

PATTERN (TEXTURE)

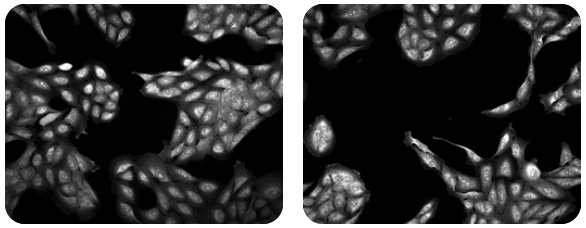



size of shift from 7 Jun, same scale

30 Aug, first flagged, 5 plates

3 Oct, latest, 14 plates

ER, BOTH LOOK FINE ON SCREEN



7 JUN

30 AUG

**7×** brighter in raw camera counts, one field each
7,612 → 51,230

one field each, ER pattern: around the nucleus → whole cell

LAB 2 (YOU), IN NUMBERS

ER + AGP brightness: up in ≥ 96% of measurements
cells per well: 189 → 165 → 182
inside its own range, 150–199, A13

CAMERA OR CELLS?

Cell measurements moved,
not only whole-image ones.
New stain batch or real biology?
This check cannot tell.

WHAT YOU GET

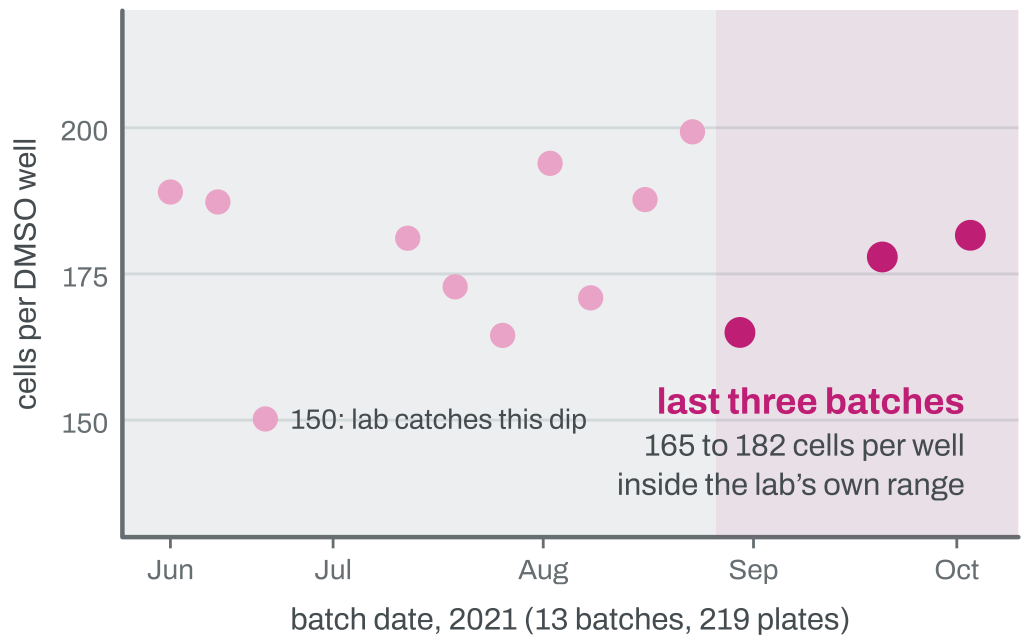
Look first: camera exposure, ER stain
Then: stain mix-up, colour leak, biology
Your lab finds the cause

ER, AGP, Mito, DNA: four stain channels, each marking different parts of the cell, each bar averages one channel’s measurements, JUMP [7]

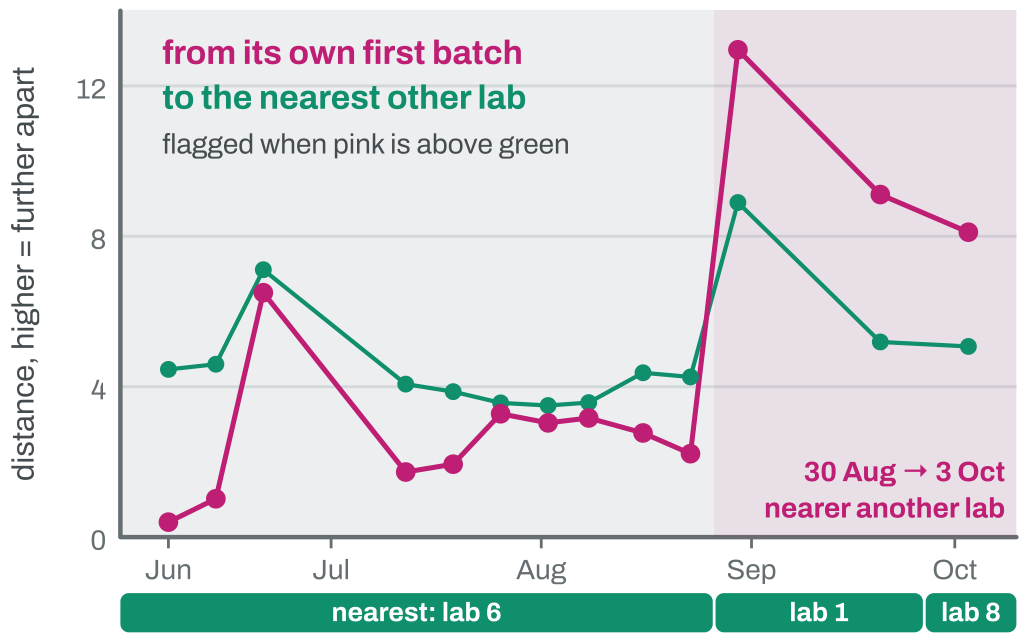
# By its own count lab 2 was fine; beside other labs, it had moved.

Lab 2 (you), 13 batches, two views.
Not a verdict: a reason to look.

LAB'S OWN CHECK: CELLS PER WELL




COMPARED WITH OTHER LABS




distance unit: usual spread between wells on one plate, lab 2, 2021, JUMP [7], which batches: A11, what moved: A12

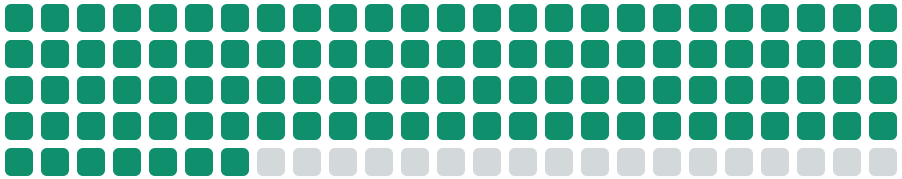
# On the map, labs agree a little more on the same drug.

Seven drugs with a known effect, 125 batches, 11 labs.
Higher is better.

## Do labs agree on what a drug does?

**0.75** raw → **0.76** usual fix → **0.78** on the map

1 = every lab sees the same effect

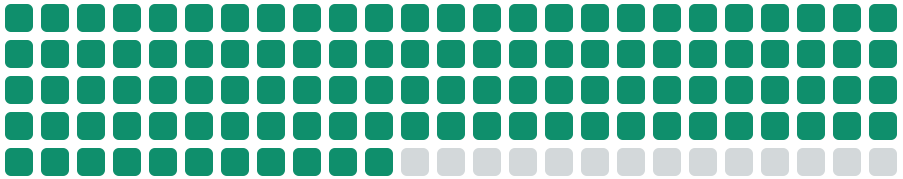


**107 of 125 batches** higher on the map than with the usual fix
than raw: 121 of 125

## Can you still tell the drugs apart?

**0.52** raw → **0.56** usual fix → **0.59** on the map

gap to the nearest wrong drug, higher is easier



**111 of 125 batches** higher on the map than with the usual fix
than raw: 125 of 125

### NOTHING LOST

Right drug recognised: 0.92 → 0.94
Matches its own past: 0.88 → 0.88

### HOLDS

Labs agree more: 5 of 5 ways of checking
Drugs stay apart: 4 of 5

### NOT FIXED YET, DRUG STRENGTH

Varies batch to batch, lower is better
raw 6% → usual fix 20% → map 12%: steadier than the usual fix, not yet as steady as raw

usual fix: each batch set against its own “normal”, its DMSO wells, map: against every other lab’s, 1,829 plates, JUMP [7]

# Labs join free; subscribers pay for access.

Certificate pays first.
Then subscriptions to the repository: access, not ownership.

| WHO | WHAT THEY GET | WHAT THEY PAY | WHAT IT BUILDS |
|---|---|---|---|
| **labs and core facilities** | a place on the map<br>certificate, leaderboard | **free, for sharing controls** | the map |
| **companies that buy cell data** | a certificate on every dataset they buy | **~$10–50k per certificate** | first revenue |
| **teams that run cell screens all year** | a check on every batch they run | **~$5–15k per cell line, per quarter** | repeat revenue |
| **AI labs and model builders** | a test set with faults planted on purpose | **~$50–200k a year, as partners** | the standard |
| **then: subscribers to the repository** | access to certified raw images from many labs | **a subscription, each lab keeps a share** | the network |

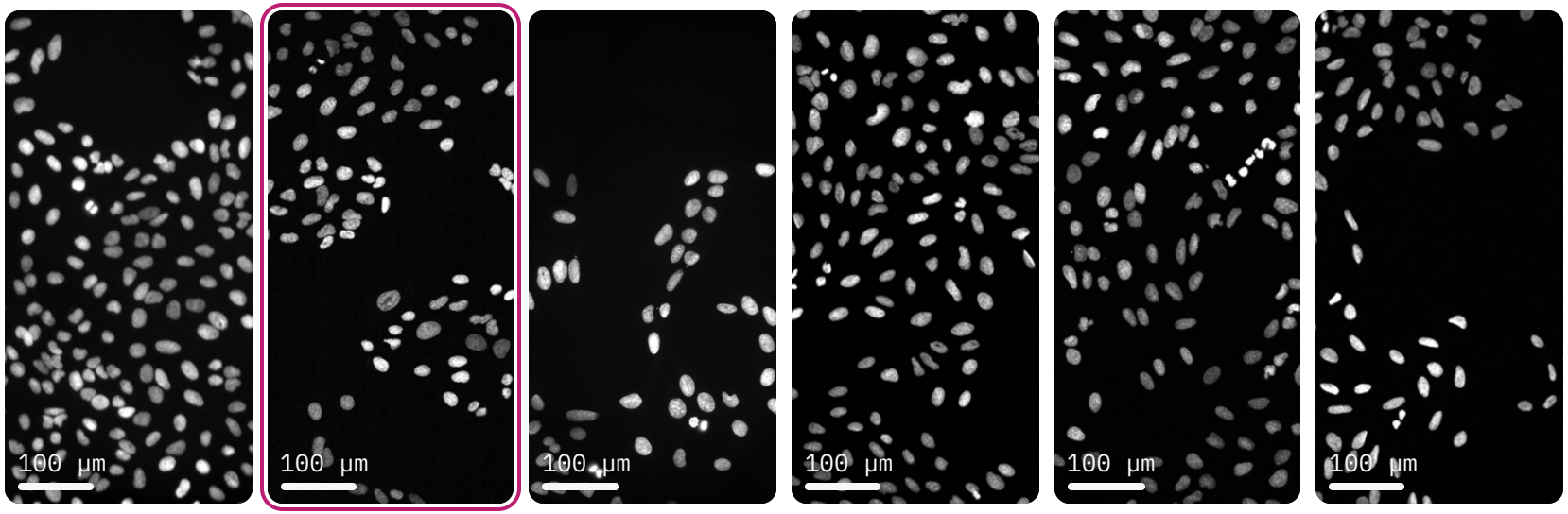
only control wells leave a lab, DMSO and known drugs, never a test compound, privacy A18, the lab’s share A24

# Biology's ugliest dataset, built on purpose.

4 cell lines × 2 microscope types × known drugs
+ planted faults. Every correction method scored.

TODAY, SAME CELLS, SAME PROTOCOL, NO DRUG

real wells, JUMP [7]




| LAB 1 | LAB 2 (YOU) | LAB 3 | LAB 5 | LAB 6 | LAB 10 |
|---|---|---|---|---|---|
| Phenix, widefield | CV8000, confocal | Phenix, widefield | CV8000, confocal | CV8000, confocal | CV8000, confocal |

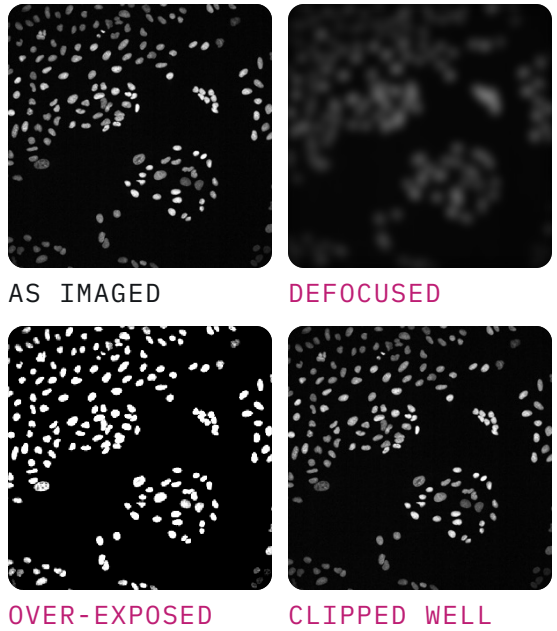
**5** microscope models

**8** plate layouts

**13** labs [7]

PLANTED FAULTS

simulated



AS IMAGED

DEFOCUSED

OVER-EXPOSED

CLIPPED WELL

Can a model name the fault?

cell lines: U2OS, A549, HepG2, iPSC [17], microscope types: widefield + confocal

# Ranked on controls, never on results.

Every lab scored on the other labs' scale, never its own.
Lab 2 (you): 8th of 11 labs.

| LAB | BATCHES | SCORE | BEST RANK | CLEAN RUN | HOW UNUSUAL | PLATES DIFFER | BATCH TO BATCH | CELL COUNT OFF |
|---|---|---|---|---|---|---|---|---|
| lab 5 | 33 | **70** | #1 | 33 | 3.35 | 0.27 | 0.81 | 0.31 |
| lab 6 | 13 | **60** | #16 | 13 | 4.44 | 0.41 | 1.23 | 0.12 |
| lab 10 | 17 | **54** | #10 | 17 | 3.84 | 0.62 | 1.21 | 0.23 |
| lab 8 | 4 | **52** | #54 | 4 | 4.10 | 1.08 | 0.86 | 0.09 |
| lab 11 | 5 | **47** | #23 | 2 | 3.79 | 0.87 | 1.57 | 0.17 |
| lab 3 | 13 | **44** | #18 | 6 | 3.27 | 0.80 | 2.01 | 0.29 |
| lab 7 | 8 | **42** | #69 | 8 | 6.38 | 0.58 | 1.17 | 0.33 |
| **lab 2** | 13 | **39** | #48 | 10 | 5.68 | 0.57 | 1.99 | 0.14 |
| lab 1 | 6 | **32** | #78 | 6 | 8.68 | 1.10 | 2.03 | 0.09 |
| lab 9 | 11 | **25** | #97 | 11 | 9.97 | 0.50 | 1.76 | 0.45 |
| lab 15 | 6 | **23** | #68 | 6 | 5.53 | 0.87 | 1.69 | 0.44 |

SCORE

## 0 to 100, higher is better

all labs: how unusual 4.01, plates differ 0.51, batch to batch 1.17, cell count off 0.28
average rank on the four measures, among 129 batches
measures: lower is better, green beats all labs

BEST BATCH

## 82

lab 5, 28 Jun, 7 plates
lab 5 holds 9 of top 10

YOUR CHOICE TO PLAY

Named only if you opt in.
Private version for your own plates.

Lab medians, 1,871 plates, 11 labs, clean run: batches in a row never flagged (A11), JUMP [7]

# Only control wells leave the lab.

No compound, no target, no result.
Nothing for legal to approve.

## WHAT LEAVES THE BUILDING

### control wells only: DMSO and known drugs



pink: 64 of 384 wells leave, JUMP [7]

### images, or numbers from them

proposed: images deleted once measured

### a plate map and the batch date

which wells, and when they were imaged

## WHAT DATUM LINE DOES WITH IT

### builds the map

one average point per plate.
Scale set by the other labs, never by yours.

### the map holds DMSO wells only

**82,319**

for lab 2: DMSO wells of 10 other labs
0 treated

### answers

where you sit, how far you moved, nearest lab, cell count, “normal” or not

### ranks every batch

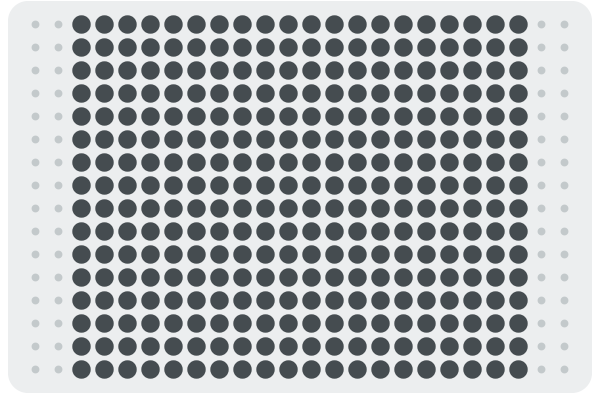
on its DMSO wells, never on results, A17

### sends a correction, not a model of you

one per batch, fitted on your DMSO wells, A19

## WHAT IT NEVER SEES

### your treated wells



dark: 320 of 384 wells stay with you

### your results

corrected in your building, by you

### your name, unless you choose

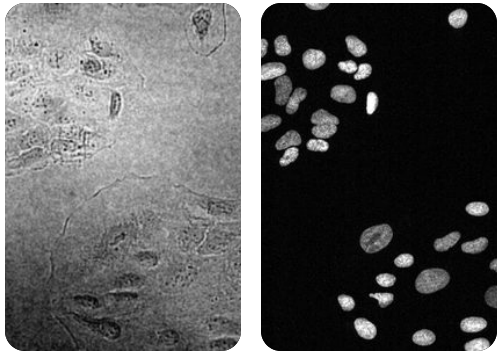
proposed: a number until you opt in.
Every lab here is a number.

In every analysis here, the map and the correction see DMSO wells only, github.com/kat-titter/datum-line/analysis, JUMP [7]

# An image goes in; a position and a correction come out.

Results stay in the lab.
DMSO wells and the correction travel.

## 1, AN IMAGE



brightfield DNA stain

one DMSO well, JUMP [7]

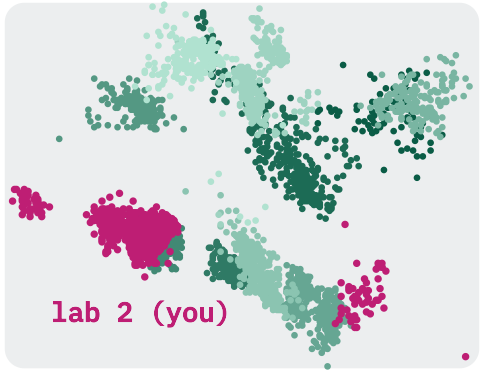
## 2, NUMBERS

**3,773**

numbers per well

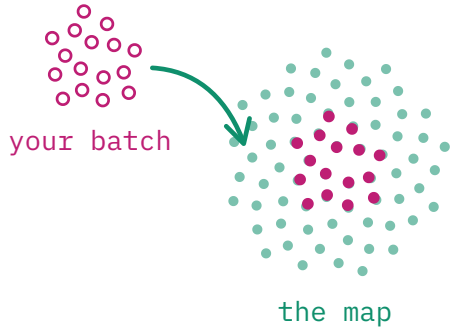
standard software today: size, shape, brightness, texture

## 3, THE MAP




each lab placed on the other labs' scale, never its own

## 4, THE CORRECTION




one per batch, fitted on your DMSO wells, sketch, A14

## EVERY LAB LOOKS DIFFERENT

**99.9%**

a computer names the lab of each well, by guessing: 9%
93,228 DMSO wells, 11 labs, on plates it never saw

## AI IMAGE MODEL NEXT, A10

brightfield, 8 labs 99.5%

DNA stain, 8 labs 96.3%

right lab, from one picture

Correction, per batch on 200 components: $x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$, every script: github.com/kat-titter/datum-line/analysis

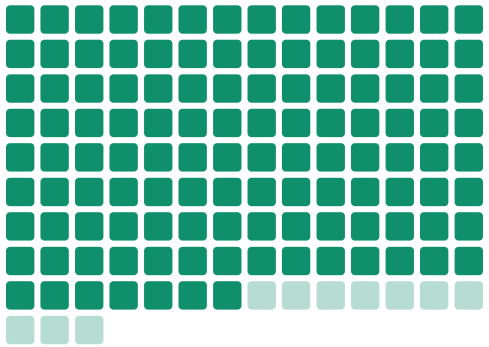
# Every lab leaves a fingerprint, so a wrong label stands out.

One map of DMSO wells, no results stored.
Two checks work today, two are proposed.

PROVENANCE today

## 99.9%

of 93,228 DMSO wells get the right lab, on unseen plates. The fingerprint is cells and microscope together: image-quality numbers alone name the lab at 99.9%.



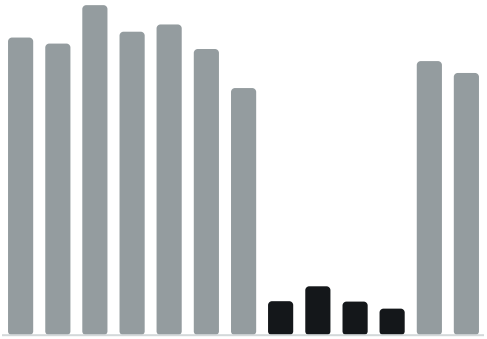
1 square = 1 batch
solid: every well right, 119

INTEGRITY today

## 4 batches

Label: known drug in these wells.

Cells: almost no effect, on 65 plates.



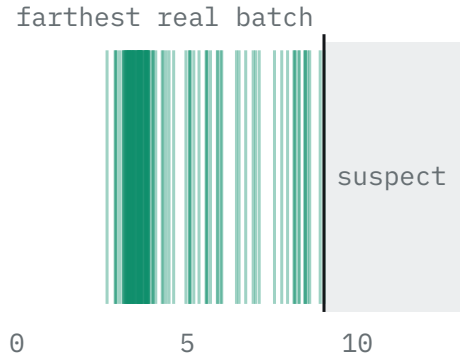
lab 3, drug effect per batch
dark: failed, A8

UNKNOWN ORIGIN proposed

## 9×

Farthest of 129 real batches from its nearest other lab.

Farther out: wrong cell line? Mixed culture? Untested.



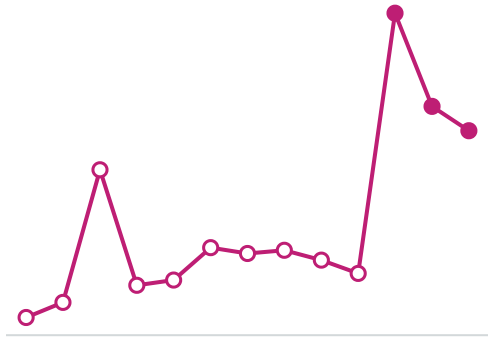

distance to nearest other lab
1 line = 1 real batch

EARLY WARNING proposed

## 30 Aug

2021: lab 2's DMSO wells moved; its cell count did not.

A move nobody explains: check reagents, air, water.



lab 2 (you), since batch 1
filled: flagged, A13

Integrity check uses each plate's known-drug wells (A8), × = times the usual spread of wells within one plate, JUMP [7]

proposed, every line an assumption

# About $1.8M for eighteen months, mostly people.

$884k planned × 2 = $1.77M asked



people 71% | benchmark 17% | other 12%

| | | |
|---|---|---|
| **leader, to run it** | 18 months | **$180k** |
| **scientific lead (me)** | 18 months | **$180k** |
| **ML engineer** | from month 7 | **$160k** |
| **payroll overhead** | 20% of salaries | **$104k** |
| **benchmark playground** | ~60 plates, planted faults, A16 | **$120k** |
| **partner-lab bench time** | 3 labs, controls they already image, reagent top-ups, shipping | **$30k** |
| **compute and storage** | images, models, hosting, API | **$45k** |
| **legal, IP, insurance, accounting** | incorporation to first contract | **$45k** |
| **travel and partner visits** | 2 benchmark partners, 3 labs | **$20k** |
| **planned** | sum | **$884k** |
| **+ buffer** | most things take 2–3× the time and money planned, we plan for 2× | $884k |
| **asked, 18 months** | | **$1.77M** |

NO IN-HOUSE LAB

Partner labs or a contract lab.
Own lab: costs more, proves less.

WHAT THE MONEY MUST SHOW

**1** benchmark playground, paid in this round **2** three partner labs on the map
**3** model for any image **4** first paid certificate → seed

salaries below market, leader and scientific lead paid, nothing here committed

# About $84M a year, if every likely buyer joined.

Count × price.
Three streams.

| assumption | low | base | high |
|---|---|---|---|
| organisations testing drugs on a few cell lines | 200 | 400 | 800 |
| cell lines each one uses per quarter | 2 | 4 | 8 |
| price per cell line per quarter, $k | 5 | 10 | 15 |
| organisations testing drugs on hundreds to thousands of cell lines | 10 | 25 | 50 |
| site licence per year, $k | 50 | 150 | 300 |
| **per-line checks and site licences, per year** | **$8.5M** | **$67.8M** | **$399M** |
| companies buying cell data they did not make | 50 | 150 | 300 |
| purchased datasets each certifies per year | 2 | 4 | 8 |
| price per certificate, $k | 10 | 25 | 50 |
| **certificates on purchased data, per year** | **$1M** | **$15M** | **$120M** |
| benchmark partners | 4 | 10 | 20 |
| membership per partner per year, $k | 50 | 100 | 200 |
| **benchmark membership, per year** | **$200k** | **$1M** | **$4M** |
| **total per year, if every buyer joined** | **$9.7M** | **$83.8M** | **$523M** |

BASE CASE, PER YEAR




↑ five-year target, a tenth, $8.4M

benchmark $1M ↑

WHY BUYERS PAY

One screen I ran on a line checked late:
3 runs a week × 17 weeks × $50k ≈ $2.6M to redo.
Certificate @ $25k: about 1% of that.
Subscription: insurance against it.

prices proposed on A15, academic labs join free, they are the map, not the revenue

# Cell data sells, but data from different labs does not add up.

Three public prices.
Same cells, four labs, different results.

AI TRAINING DATA

## $1.5M a day

paid to experts for AI training data, by one vendor

~$85 an hour, $10B valuation [10]

CELL MAPS

## $30M

paid by Roche for one cell dataset

second milestone, $213M to date [11]

CONSORTIA

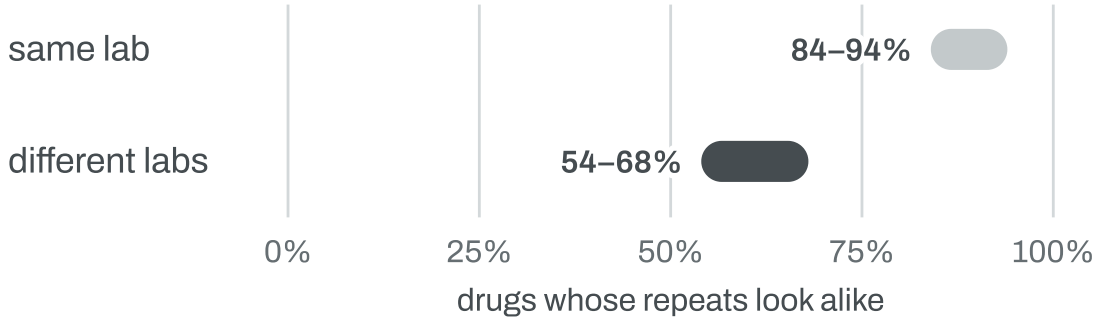
## $1.8B

pledged for cell data to train AI
funders get it a year early

Meta, DeepMind, Isomorphic: $300M [12]

4 EUROPEAN LABS, ONE PROTOCOL, ONE CELL LINE

same cells, same reagents, shipped to every lab [13]




WHAT THE PAPER SAYS

**“not directly comparable between imaging sites”** [13]

ON ONE SHARED MAP, JUMP [7]

## 107 of 125

batches: labs agree more on what a drug does, A14

[7] JUMP, 2023, [10] TechCrunch, 2025, [11] Recursion, 2025, [12] eWeek, 2026, [13] Wolff et al., *iScience*, 2025, full references in the README

# The lab keeps its data and a share; anyone can check it.

Subscribers get access, not ownership.
No crypto token.

### CONTROL WELLS ONLY — today

no test compound, no target, no result
results never leave the lab

### LAB KEEPS ITS DATA — proposed

sharing is opt-in
withdraw any time

### LOW RISK BY DESIGN — today

cell-line images only
no human genetic data, no pathogen data

### LABS LEAVE FINGERPRINTS — today

**99.9%** right lab out of 11
from control-well images A20

### SIGNED AND TIMESTAMPED — proposed

public record, anyone can check
dates cannot be changed, blockchain optional

### KNOWN SUBSCRIBERS — proposed

verified at sign-up, odd access flagged
access log open to members

**lab shares**
images of control cells
certificate attached

→

**certificate checks**
how “normal” the cells look
which lab, what changed

→

**subscriber pays**
to use certified data
from many labs

→

**lab keeps a share**
of every subscription
as long as its data is used